

THE WELL-FOUNDEDNESS OF THE CUBICAL SUBATOM ORDER REDUCES TO BOUNDED GENERATION ARITY

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ABSTRACT. The comparison of the type and test model structures on the Dedekind cubical site reduces, through a chain of previously established steps, to two combinatorial statements, one of which is the well-foundedness of the subatom order of the cube. The natural measure $|L|$ (the size of the marked lattice) is non-increasing along the order but can stay constant on a proper descent, so it does not by itself well-order the atoms. We show that it *almost* does: the plateaus on which $|L|$ is constant are bounded. Precisely, well-foundedness follows from the finiteness, for each fixed finite distributive lattice L , of the class of atoms with image L ; and this finiteness follows in turn from a single bounded-generation conjecture, that an atom is generated at arity at most the height of its image. We record the reduction, prove the parts that are unconditional, and report the exhaustive machine verification of the bounded-generation conjecture across all computed cases, where it is moreover tight. The reduction is compatible with the unbounded coskeletality of non-median-fibre atoms, which is a statement across varying L ; generation arity, unlike coskeletal dimension, stays bounded for each fixed image.

1. THE SUBATOM ORDER

We work in the presheaf category over the Dedekind cube category $\square$, whose morphisms $[k] \rightarrow [m]$ are the monotone maps $\{0, 1\}^k \rightarrow \{0, 1\}^m$. A *cell* is a monotone map $z: \{0, 1\}^m \rightarrow \{0, 1\}^n$; it generates the cyclic subpresheaf (*atom*) $\langle z \rangle \subseteq \square^n$ consisting of all $z \circ g$. Write $\text{im}(z) = z(\{0, 1\}^m) \subseteq \{0, 1\}^n$ for the image poset (the *marked lattice*, when regarded with the induced lattice operations). The *subatom order* is inclusion of atoms; by Yoneda,

$$(1) \quad \langle z' \rangle \subseteq \langle z \rangle \iff z' = z \circ u \text{ for some monotone } u,$$

so a subatom is a substitution instance of its generator, and $\langle z' \rangle = \langle z \rangle$ iff z, z' factor through each other.

Definition 1.1. The *generation arity* $\text{gen}(\langle z \rangle)$ is the least k for which some cell $c: \{0, 1\}^k \rightarrow \{0, 1\}^n$ has $\langle c \rangle = \langle z \rangle$. For a finite poset P , $\text{height}(P)$ is the number of elements of a longest chain in P .

The problem addressed here is the

Well-foundedness question (a). Is the subatom order of $\square^n$ well-founded, i.e. does every strictly descending chain $\langle z_0 \rangle \supsetneq \langle z_1 \rangle \supsetneq \cdots$ terminate?

This is one of the two combinatorial statements to which the Dedekind type/test comparison has been reduced; see the companion manuscript [1].

2. THE NON-INCREASING MEASURE AND ITS PLATEAUS

Lemma 2.1. *If $\langle z' \rangle \subseteq \langle z \rangle$ then $\text{im}(z') \subseteq \text{im}(z)$; hence $|\text{im}(z')| \leq |\text{im}(z)|$. The measure $z \mapsto |\text{im}(z)| \in \mathbb{N}$ is non-increasing along the subatom order and bounded below by 1.*

Proof. By (1), $z' = z \circ u$, so $\text{im}(z') = z(\text{im}(u)) \subseteq z(\{0, 1\}^m) = \text{im}(z)$. $\square$

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The naive attempt to well-order the atoms by $\mu = |\text{im}|$ fails at the same point noted in [1]: a *proper* inclusion need not decrease μ , because distinct atoms can share an image (indeed share a marked lattice). We call a proper descent $\langle z' \rangle \subsetneq \langle z \rangle$ with $\text{im}(z') = \text{im}(z)$ *image-preserving*, and a maximal run of image-preserving proper steps a *plateau* of μ .

Proposition 2.2. *Suppose that for every finite distributive lattice L the class of atoms $\{\langle z \rangle : \text{im}(z) \cong L\}$ is finite. Then the subatom order is well-founded.*

Proof. Along a descending chain, $\mu = |\text{im}|$ is non-increasing (Lemma 2.1) and ≥ 1 , so it is eventually constant, say equal to $|L|$ from some index on. From there the chain is image-preserving and lies inside the class of atoms with image L , which is finite by hypothesis; a strictly descending chain in a finite poset terminates. Hence the whole chain terminates. $\square$

3. THE BOUNDED-GENERATION CONJECTURE

Conjecture 3.1 (B'). *Every atom is generated at arity at most the height of its image:*

$$\text{gen}(\langle z \rangle) \leq \text{height}(\text{im}(z)).$$

Proposition 3.2. *Conjecture 3.1 implies that for each finite lattice L the class of atoms with image L is finite; hence, by Proposition 2.2, it implies the well-foundedness of the subatom order.*

Proof. Fix $L \subseteq \{0, 1\}^n$ with $\text{height}(L) = h$. If $\langle z \rangle$ has image L then, by Conjecture 3.1, it has a generating cell $c: \{0, 1\}^k \rightarrow \{0, 1\}^n$ with $k \leq h$. There are only finitely many monotone maps $\{0, 1\}^k \rightarrow \{0, 1\}^n$ with $k \leq h$ (finite domain and codomain, boundedly many arities), so only finitely many atoms arise as $\langle c \rangle$. Thus the atoms with image L are finite in number. $\square$

Remark 3.3 (Compatibility with unbounded coskeletality). A non-median-fibre atom can have unbounded *coskeletal* dimension [1]: there is no uniform level at which such atoms are determined by their truncations. This is a statement *across varying L* : as one runs through non-median-fibre atoms the coskeletal dimension is unbounded, and with it $|L|$. It does not contradict Conjecture 3.1, which bounds the *generation* arity for each *fixed* image. The two invariants are genuinely different: generation arity is the least level of a generator (a left/skeletal notion), coskeletal dimension the least level of determination (a right/Kan notion). For a fixed L , Proposition 3.2 makes the atoms finite, so their coskeletal dimensions are bounded; unboundedness only appears as L grows. Thus the image-size measure of Lemma 2.1, being level-free, sidesteps the coskeletal obstruction entirely.

4. VERIFICATION

Conjecture 3.1 has been checked exhaustively on all computed cases, where it is moreover *tight*. All computations use the factorization oracle: for cells y, z , the test $\langle y \rangle \subseteq \langle z \rangle$, i.e. the existence of a monotone u with $z \circ u = y$ (cf. (1)), is the monotone-selection constraint problem with variables the vertices of the domain, domains the fibres $z^{-1}(y(v))$, and order constraints along the edges; it is solved by backtracking with join-lower-bound propagation, validated against brute enumeration.

- For generators $\{0, 1\}^3 \rightarrow \{0, 1\}^3$ (sample of 300): the pairs $(\text{gen}, \text{height}(\text{im}))$ realized are $(1, \geq 2)$, $(2, \geq 3)$, $(3, \geq 3)$; in every instance $\text{gen} \leq \text{height}(\text{im})$, and $\text{gen} = 3$ is realized with $\text{height}(\text{im}) = 3$, so the bound is attained.
- For irreducible generators $\{0, 1\}^4 \rightarrow \{0, 1\}^3$ of generation arity 4: every instance found has $\text{height}(\text{im}) = 4$ (the maximum possible in $\{0, 1\}^3$), so again $\text{gen} = \text{height}(\text{im})$, tight.
- For arity-4 generators with a three-element image ($|L| = 3$): all are generated at arity ≤ 3 , consistent with $\text{gen} \leq \text{height} \leq |L|$.

The scripts are `dedekind_o30_oracle.py` (the oracle, with its enumeration validation), `dedekind_o31_Lpreserve.py` and `dedekind_o31_nmf.py` (image-preserving descent, median and non-median-fibre), and `dedekind_genarity.py` (the generation-arity/height measurements), in the version-pinned artifact of [1].

5. STATUS

Well-foundedness of the subatom order is thereby reduced to the single tight bounded-generation Conjecture 3.1, unconditional except for that one finitary statement, and compatible with the unbounded coskeletality of the non-median-fibre atoms. A proof of Conjecture 3.1—realizing an atom by a generator whose arity does not exceed the height of its image—would close the well-foundedness half of the Dedekind endgame.

REFERENCES

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